PHYSICS ONLY

5 THE TURNING EFFECT OF FORCES

A force can have a turning effect – it can make an object turn around a fixed pivot point. When the anticlockwise turning effects of forces are balanced by turning forces in the clockwise direction, the object will not turn – it is in balance.

a



b



▲ Figure 5.1 Turning effects are used in many places, such as in **a** the see-saw, and **b** the cranes.

LEARNING OBJECTIVES

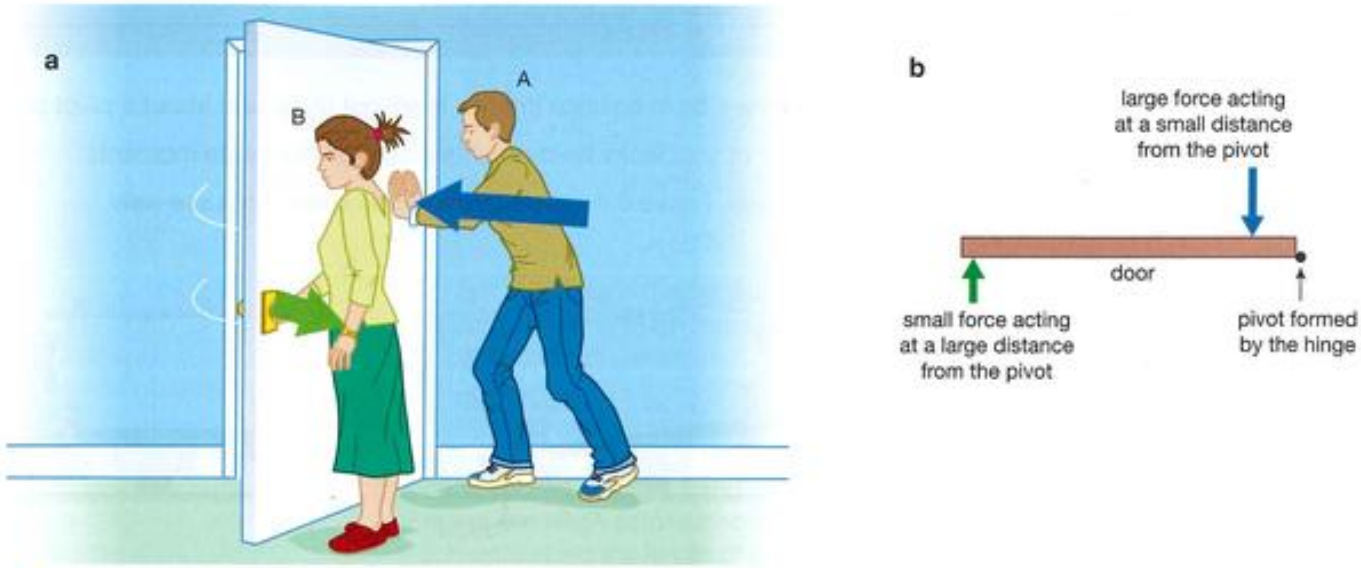
- Know and use the relationship between the moment of a force and its perpendicular distance from the pivot:

  moment = force × perpendicular distance from the pivot
- Know that the weight of a body acts through its centre of gravity
- Use the principle of moments for a simple system of parallel forces acting in one plane
- Understand how the upward forces on a light beam, supported at its ends, vary with the position of a heavy object placed on the beam

Unbalanced forces acting on objects can make them accelerate or decelerate. In the examples in Figure 5.1, the forces acting are having a turning effect. They are making the objects, like the see-saw, turn around a fixed point called a **pivot** or fulcrum. The turning effect in cranes is explained in Figure 5.6 on page 51. We use this turning effect of forces all the time. In our bodies the forces of our muscles make parts of our bodies turn around joints like our elbows or knees. When you turn a door handle, open a door or remove the lid off a tin of paint with a knife you are using the turning effect of forces. Understanding the turning effect of forces is important. Sometimes we want things to turn or rotate – the see-saw wouldn't be much fun if it didn't. However, sometimes we want the turning effects to balance so that things don't turn – it would be terrible if the crane did not balance!

OPENING A DOOR

Challenge a partner (perhaps one who thinks that he or she is strong!) to try to hold a door closed while you try to open it – then explain the rules! They can apply a pushing force but no further than 20 cm from the hinge (the part of the door that fastens it to the wall), while you try to pull the door open by pulling on the handle. You will be able to open the door quite easily.



▲ Figure 5.2 The distance of the force from the pivot is crucial.

You will realise that you have an advantage because the turning effect of your force doesn't just depend on the size of the force you apply but also on the distance from the hinge or pivot at which you apply it. You have the advantage of greater leverage (Figure 5.2).

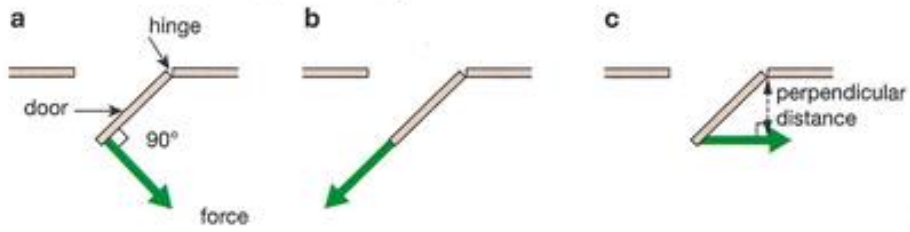
THE MOMENT OF A FORCE

The turning effect of a force about a hinge or pivot is called its moment. The moment of a force is defined like this:

moment of a force (Nm) = force, F (N) × perpendicular distance from pivot, d (m)

$$\text{moment} = F \times d$$

The moment of a force is measured in newton metres (Nm) because force is measured in newtons and the distance to the pivot is in metres. We need to be precise about what distance we measure when calculating the moment of a force. Look at the diagrams in Figure 5.3.



▲ Figure 5.3 For a force to have its biggest effect it should be at 90° to the lever.

If you think about the simple door opening 'competition' we discussed earlier you will realise that for a force to have the biggest turning effect it should be applied as in Figure 5.3a – that is, its line of action should be perpendicular (at 90°) to the door.

In Figure 5.3b the force has no turning effect at all because the line along which the force is acting passes through the pivot.

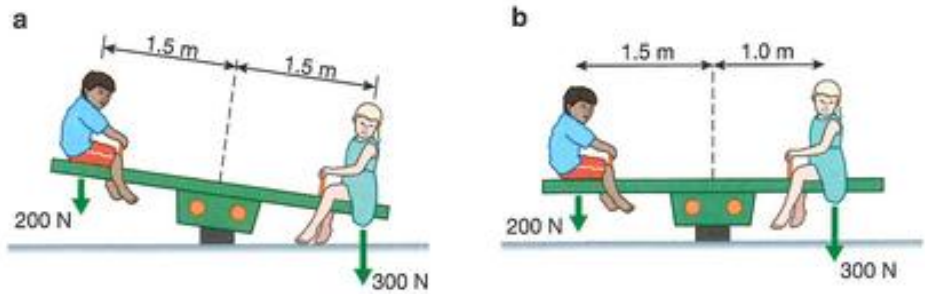
Figure 5.3c shows how the distance to the pivot must be measured to get the correct value for the moment. The distance is the perpendicular distance from the line of action of the force to the pivot.

IN BALANCE

An object will be in balance (that is, it will not try to turn about a pivot point) if:

sum of anticlockwise moments = sum of clockwise moments

For example, Figure 5.4 shows two children sitting on a see-saw.



▲ Figure 5.4 The moment of the heavier child is reduced when she sits closer to the pivot, so that it balances the moment of the lighter child.

In Figure 5.4a:

the anticlockwise moment = 200 N × 1.5 m = 300 Nm

the clockwise moment = 300 N × 1.5 m = 450 Nm

So the see-saw is not balanced and leans down to the right as it rotates clockwise about the pivot.

The calculations for Figure 5.4a are approximate because the forces are not acting perpendicularly to the see-saw.

In Figure 5.4b:

the anticlockwise moment = 200 N × 1.5 m = 300 Nm

the clockwise moment = 300 N × 1.0 m = 300 Nm

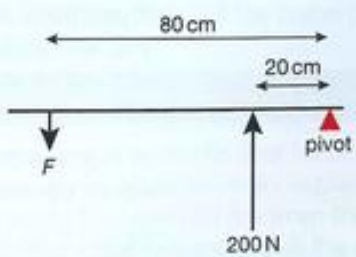
So the see-saw is now balanced.

EXAMPLE 1

Look again at Figure 5.2. Person A pushes the door with a force of 200 N at a distance of 20 cm from the hinge (the pivot in this example). Person B opens the door by pulling the door handle which is 80 cm from the hinge.

What is the minimum pulling force that person B must use to open the door?

Here is a simplified sketch of the problem:



▲ Figure 5.5 A labelled sketch of the problem

The door will be in balance (on the point of opening) if:

sum of anticlockwise moments = sum of clockwise moments

$$F \times 80\ \text{cm} = 200\ \text{N} \times 20\ \text{cm}$$

$$F = \frac{200\ \text{N} \times 20\ \text{cm}}{80\ \text{cm}}$$

$$= 50\ \text{N}$$

But this means the turning effects are balanced and we want person B to apply a big enough force to make the anticlockwise moment bigger than the clockwise moment.

The answer is that person B must apply a force greater than 50 N ($F > 50$ N).

Look back at Figure 5.1 and look at Figure 5.6. The load arm is long so that the crane can reach across a construction site and move loads backwards and forwards along the length of the arm. The weight of the long load arm and the load must be counterbalanced by the large concrete blocks at the end of the short arm that projects out behind the crane controller's cabin. The counterbalance weights must be large because they are positioned closer to the pivot point, where the crane tower supports the crosspiece of the crane. Without careful balance the turning forces on the support tower could cause it to bend and collapse.

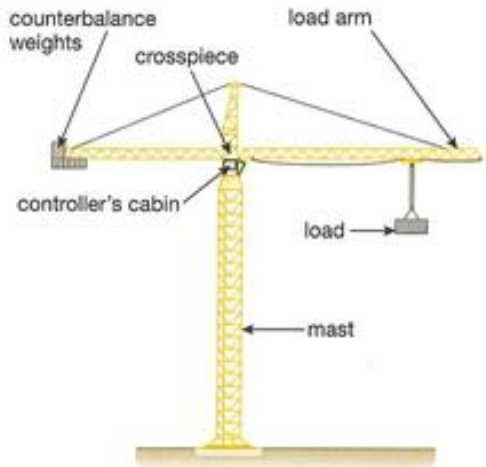


Figure 5.6 The clockwise and anticlockwise moments on the crane must be balanced.

CENTRE OF GRAVITY



▲ Figure 5.7 Can you find the point at which the ruler balances?

Try balancing a ruler on your finger, as shown in Figure 5.7.

When the ruler is balanced, the anticlockwise moment is equal to the clockwise moment, but there are no downward forces acting in this situation other than the weight of the ruler itself. We know that the weight of the ruler is due to the pull of the Earth's gravity on the mass of the ruler. The mass of the ruler is equally spread throughout its length. It is not, therefore, surprising to find that the ruler balances at its centre point.

We say that the centre of gravity of the ruler is at this point. The weight force of the ruler acts through the centre of gravity – it is the point where the whole of the weight of the ruler appears to act. If we support the ruler at this point there is no turning moment in any direction about the point, and it balances. The centre of gravity is sometimes called the centre of mass.

EXTENSION WORK

Finding the centre of gravity of a sheet of card

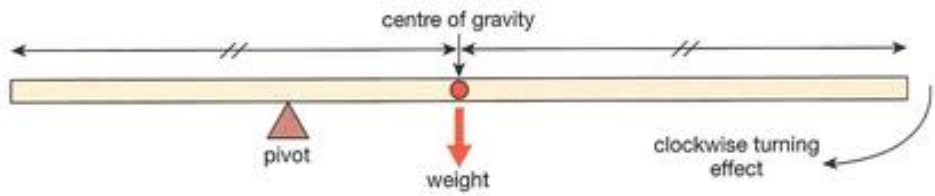
If you have a sheet of card or any other uniform material then finding its centre of gravity is quite straightforward – it will be located where the axes of symmetry cross, as shown in Figure 5.8.



▲ Figure 5.8 The centre of gravity for these three regular shapes is where the axes of symmetry cross. An axis of symmetry can be found using a plane mirror. If you place a plane mirror along one of the dotted lines in any of the above shapes, the reflection in the mirror looks exactly like the original.

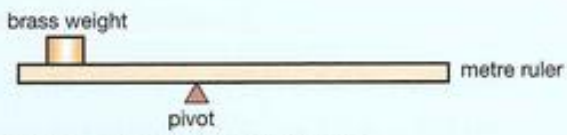
OBJECTS NOT PIVOTED AT THE CENTRE OF GRAVITY

A simple see-saw is a uniform beam (plank) pivoted in the middle. The centre of gravity of a uniform beam is in the middle, so the see-saw is pivoted through its centre of gravity. When an object is not pivoted through its centre of gravity the weight of the object will produce a turning effect. This is shown in Figure 5.9.



▲ Figure 5.9 The weight of the beam causes a clockwise turning effect about the pivot.

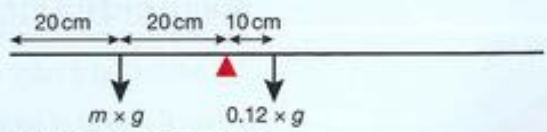
EXAMPLE 2



▲ Figure 5.10 In balance – not tipping in either direction

This sketch diagram shows a uniform wooden metre ruler with a mass of 0.12 kg. A pivot at the 40 cm mark supports the rule. When a brass weight is placed at the 20 cm mark the ruler just balances in the horizontal position. Take moments about the pivot to calculate the mass of the brass weight.

The weight of the ruler acts vertically through the centre of gravity of the rule. As the ruler is uniform the centre of gravity is at the 50 cm mark.



▲ Figure 5.11 A labelled sketch of the problem

For balance:

anticlockwise moments = clockwise moments

$(m \times g) \times 20\ \text{cm} = (0.12\ \text{kg} \times g) \times 10\ \text{cm}$

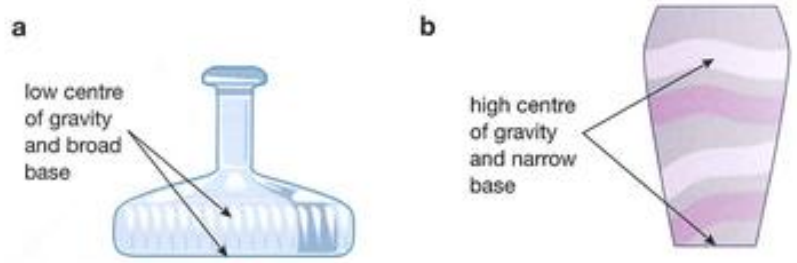
The mass of the brass weight is 0.06 kg.

Here g is the gravitational field strength in N/kg but, as you can see, it cancels. You can leave the distances in centimetres because you are using the same length units on both sides of the equation.

EXTENSION WORK

Centre of gravity and stability

The position of the centre of gravity of an object will affect its stability. A stable object is one that is difficult to push over – when pushed and then released it will tend to return to its original position.



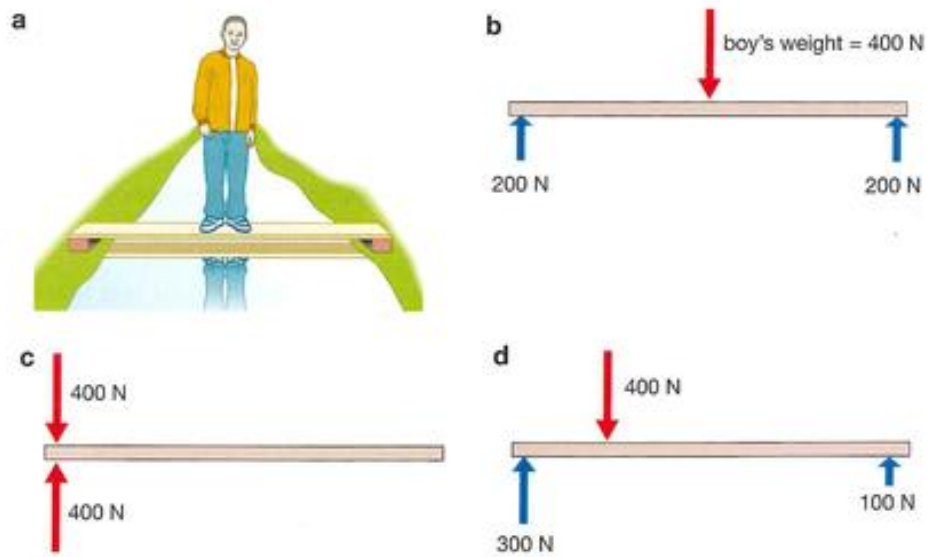
▲ Figure 5.12 Stable and **unstable** objects

The shape in Figure 5.12a is typical for a ship's decanter (bottle). Bottles used on ships need to be difficult to knock over for obvious reasons! It is stable because, when knocked, its low centre of gravity and wide base result in a turning moment that tries to pull it back to its original position. The object in Figure 5.12b has a higher centre of gravity and smaller base, so it is much less stable. Only a small displacement is needed to make it fall over.

FORCES ON A BEAM

Figure 5.13a shows a boy standing on a beam across a stream. The beam is not moving, so the upward and downward forces must be balanced. As the boy is standing in the middle of the beam, the upward forces on the ends of the beam are the same as each other. The forces are shown in Figure 5.13b – to keep things simple, we are ignoring the weight of the beam.

If he moves to one end of the beam, as shown in Figure 5.13c, then the upward force will all be at that end of the beam. As he moves along the beam, the upward forces at the ends of the beam change. In Figure 5.13d, he is one-quarter along the beam. The upward force on the support nearest to him is three-quarters of his weight, and the upward force on the end furthest away from him is only one-quarter of his weight.



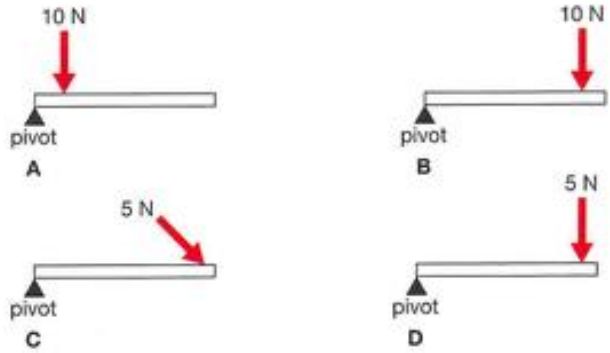
▲ Figure 5.13 The forces at the ends of a beam depend on where the load is applied.

CHAPTER QUESTIONS

More questions on the turning effect of forces can be found at the end of Unit 1 on page 55.

SKILLS ANALYSIS

1 Look at the diagram below. It shows various forces acting on objects about pivots.

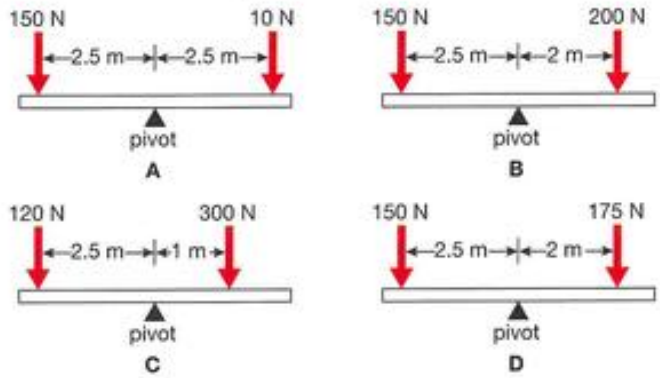


Write down the letter of the diagram that:

a shows the greatest turning moment about the pivot

b shows a turning moment of zero.

2 Look at the four see-saw diagrams below.



Write down the letter(s) of the diagram(s) that agree with the following statements, or write 'none' if no diagram agrees with the statement.

a The see-saw is balanced.

b The see-saw will rotate clockwise (dip down to the right).

c The see-saw will rotate anti-clockwise (dip down to the left).

SKILLS INTERPRETATION

3 A bookshelf is 2 m long, with supports at its ends (P and Q). Do not consider the weight of the bookshelf when you are answering these questions.

a Draw a sketch of the shelf, showing the supports.

SKILLS PROBLEM SOLVING

b A book weighing 10 N is placed in the middle of the shelf. What are the upward forces at P and Q?

c The book is moved so that it is 50 cm from Q. Use moments to calculate the forces at P and Q.

d The bookshelf weighs 10 N. Repeat parts b and c taking into account the weight of the shelf as well as the weight of the book.

END OF PHYSICS ONLY